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Tca4rec: contrastive learning with popularity-aware asymmetric augmentation for robust sequential recommendation

Yanan Bai[1], Liji Xiao[1], Chongjun Xia[2], Kexiang Zeng[2] and Xiaoyu Shi[2*]

*Correspondence: xiaoyushi@cigit.ac.cn

[1] School of Artificial Intelligence, Chongqing University of Technology, Chongqing 401135, China
[2] Chongqing Institute of Green and Intelligent Technology, Chinese Academy of Sciences, Chongqing 400714, China

Abstract

Sequential recommender systems play a pivotal role in modern recommendation scenarios by capturing users' dynamic interests through their historical interactions. While existing methods often rely on sophisticated deep models to enhance recommendation quality, they suffer from performance degradation due to sparse supervision signals and popularity bias in the training data. In this paper, we propose TCA4Rec, a robust sequential recommendation framework that addresses these challenges via a novel two-stage contrastive learning approach. Our framework incorporates an additional memory module to aggregate sequence embeddings, thereby providing flexible and generalized representations of user preferences. To mitigate popularity bias, we derive an Asymmetric Multi-instance Noise Contrastive Estimation (AMINCE) loss function that supplies rich, bias-aware training signals, while our two-stage training strategy significantly reduces the over-dominance of popular items during optimization. Extensive experiments are conducted on three real-world datasets. The experimental results demonstrate that TCA4Rec achieves significant improvements over state-of-the-art baselines, in terms of recommendation accuracy and robustness against sparse and noisy data. It attains relatively gains of 19.26% in HR@5 and 17.97% in NDCG@5 on the Amazon-sports dataset. The code is available at https://github.com/shixiaoyu0216/TCA4Rec/tree/main.

Keywords: Sequential recommender system, Robust recommendation, Contrastive learning, Popularity bias, Data sparsity

Introduction

Recommender systems (RS) play an essential role in fulfilling user needs on online platforms such as Amazon, Alibaba, and TikTok. In these environments, user preferences are inherently dynamic and evolve over time, presenting a significant challenge for accurate recommendation generation. To address this challenge, various sequential recommendation methods have been developed to enhance RS by capturing the evolving preferences of users through their sequential interaction records [1–5]. In recent years, sequential recommender (SR) systems have become a critical component of modern RS.

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The core procedure of sequential recommendation is to learn the complex sequential patterns of user behaviors. Significant efforts have been directed towards sophisticated deep learning models to achieve it [6–12]. These methods include leveraging Recurrent Neural Networks (RNN) [13] and Self-attention mechanisms [14] to capture complex sequential patterns in user behavior sequences, as well as utilizing Graph Neural Networks (GNN) to explore intricate item transition patterns within user sequences [15, 16].

Despite the promising results of these methods, they still face challenges in recommendation robustness:

- Recent studies [4, 17] have revealed that popularity bias constitutes a fundamental challenge in RS. Popularity bias has been empirically verified by Abdollahpouri et al. [18]. It can be defined as follows:

Definition 1 Popularity bias refers to the problem where popular items are recommended even more frequently than their popularity would warrant, amplifying long-tail effects already present in many recommendation domains [17].

As we all know, the long-tail effect is a well-known phenomenon in the RS domain: a small number of popular items occupy a great number of user interactions, while the majority of items receive little attention from the users. Based on such long-tail data, the trained recommendation algorithm prefers giving a higher score to popular items than their ideal values. Following [19], the popularity of an item i is defined as the number of clicked (i.e., frequency) on item i. Figure 1 shows the popularity bias effects in the dataset Amazon. We can observe that 20% of items receive 58.1% click counts from users on Amazon-Beauty. The same phoneomna also happend on dataset Amazon-Sport, 20% of items receive 56.8% click counts form users. By using such data for training, these popular items enjoy a higher degree of recommendation frequency. As a result, recommenders trained on such skewed data tend to push the items with high popularity for recommendation, leading to the infamous Matthew effect.

- In RS, the primary objective directly related to the recommendation is still the next item prediction, commonly implemented by cross-entropy loss or Bayesian Person-

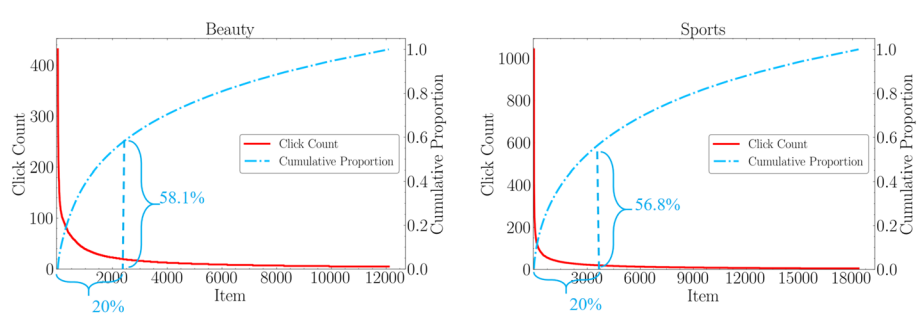

Fig. 1 The popularity bias effects in dataset Amazon in terms of Items Vs Click Count & Cumulative proportion. Left is the result on dataset Amazon-Beauty and the right one is the result on dataset Amazon-Sport

alized Ranking (BPR). Most current methods heavily emphasize the quality of the supervision signal, assuming that user interaction sequences are clean, i.e., free from noisy and irrelevant interactions. However, the supervision signal is often too sparse or noisy to effectively train a model.

To address the challenges of sequential recommendation, an emerging solution involves integrating auxiliary tasks that enable models to learn from diverse supervision signals. In recent years, self-supervised learning techniques have demonstrated remarkable success in representation learning across natural language processing (NLP) and computer vision (CV), by leveraging intrinsic data correlations from unlabeled data. Inspired by these advancements, contrastive learning (CL) has been increasingly adopted to enhance sequential recommendation performance. By optimizing a contrastive objective on different views of the same sample or semantically similar samples, CL-based approaches significantly improve representation quality by pulling positive samples closer while pushing negative samples apart. Despite these improvements, existing CL-based sequential recommendation models suffer from two major limitations. First, due to the imbalanced data distribution, positive and negative samples are often drawn from a long-tailed distribution where popular items have a disproportionately high sampling probability. This leads to models that overly emphasize popular items while neglecting long-tail content. Second, the inherent sparsity of supervision signals in sequential data further degrades model performance, as popular items dominate the learning process and skew the overall recommendations.

To overcome these limitations, we propose TCA4Rec, a novel Two-stage Contrastive Learning framework for Sequential Recommendation. Our approach introduces three key components: (1) TCA4Rec first incorporates an additional memory module to aggregate sequence embeddings, which provides flexible and generalized representations of user preferences while capturing long-term dependencies; (2) To mitigate popularity bias, TAC4Rec introduces an Asymmetric Multi-instance Noise Contrastive Estimation (AMINCE) loss that generates asymmetric positive and negative samples by balancing popular and non-popular items. Unlike conventional contrastive learning methods, AMINCE supplies rich, bias-aware training signals that better represent the long-tail distribution; (3) TAC4Rec explicitly divides the training process into two stages. **Stage 1** focuses on contrastive representation learning with AMINCE to establish robust item embeddings. **Stage 2** employs a selective update strategy, where only a random subset of popular items is updated in each iteration, while long-tail items are continuously refined. This strategy ensures that the training process does not become dominated by popular items. Through extensive experiments on three real-world datasets, TCA4Rec significantly outperforms state-of-the-art baselines, yielding superior performance in both hit rate and ranking quality. Moreover, its effectiveness in mitigating popularity bias makes it well-suited for practical applications such as e-commerce, video streaming, and computational advertising.

The contributions of this paper are summarized as follows:

- We develop the TCA4Rec framework for robust sequential recommendation, which jointly alleviates the sparse supervision signals and the popularity bias of CL-based methods. This is an important aspect that has been overlooked in current work.

- We derive an Asymmetric Multi-instance Contrastive Estimation loss, denoted as AMINCE, for the sequential recommendation task in batch training mode, which provides rich bias-aware training signals aligned with the recommendation. Additionally, a two-stage training strategy is proposed to further address the issue of popular items dominating model training.
- We conduct experiments on three real-world recommendation datasets to demonstrate the superiority and effectiveness of the proposed TCA4Rec framework in terms of recommendation accuracy and robustness, compared to state-of-the-art baselines

The remainder of this paper is organized as follows. Section "Related work and background" summarizes the related work. Section "TCA4Rec" gives the problem statement formally and describes the proposed TCA4REC framework in detail. Section "Experiments" presents the experiment details. Section "Conclusion and future Work" concludes our paper.

Related work and background

In the following sections, we first review the previous works on sequential recommendation and contrastive learning approaches for recommender system, then introduce the prerequisites about single-instance augmentation methods.

Related work

Sequential recommendation

User shopping behaviors typically manifest in a sequential manner rather than in isolation. Therefore, accurately capturing users' dynamic preferences based on their historical interactions is crucial for effective recommendations. Early works on sequential recommendation often relied on the Markov Chain (MC) assumption to model sequential patterns. For instance, Rendle et al. [20] combined first-order MC with matrix factorization, achieving promising results in predicting subsequent actions. To capture more complex patterns by considering multiple preceding items, high-order MCs were later explored [21]. With the advent of deep learning, Recurrent Neural Networks (RNNs) and their variants, such as Long Short-Term Memory (LSTM) [22] and Gated Recurrent Units (GRU) [13], have been widely adopted for sequential recommendation tasks. For example, GRU4Rec [13] first utilizes the RNN network to learn the sequential patterns. Quadrana et al. [23] propose a hierarchical RNN network to capture the cross-session dependencies in user behavior sequences. However, high-order Markov chain models incorporate limited historical information, while RNN-based methods are predicated on the strict order assumption.

To address the shortcomings of MC-based and RNN-based models, several attention-based RNN methods have been proposed to handle noise or irrelevant interactions in user behavior sequences [14, 24–27]. To model union-level and point-level sequential patterns, Caser [28] utilizes the convolutional filter with horizontal and vertical sliding window strategies to capture short-term contexts for prediction. TiSASRec [29] leverages a self-attention model for sequential recommendation, which can adaptively assign varying weights to different items by considering the absolute position and time interval

in a sequence. Parallel to these developments, Graph Neural Networks (GNNs) have gained considerable attention for their ability to capture complex item transition patterns within user sequences, particularly when these sequences can be represented as graphs. For instance, Wu et al. [30] employed gated graph networks to aggregate neighborhood information in directed graphs, while Qiu et al. [31] leveraged attention mechanisms for more nuanced aggregation, differentiating the influence of various neighbors.

Most current methods heavily emphasize the quality of the supervision signal, assuming that user interaction sequences are clean, i.e., free from noisy and irrelevant interactions. However, the supervision signal is often too sparse or noisy to effectively train a model. To address this, hybrid approaches integrating trust information and clustering techniques have been explored to mitigate data sparsity and cold-start problems [32, 33]. Moreover, a scalable and data-independent multi-agent recommender system using social network analysis has been proposed to address these challenges by leveraging community detection and link prediction techniques [34].

Hybrid AI-Driven Optimization in Sequential Recommendation. Recent advances in AI-driven optimization techniques have paved the way for hybrid models that further enhance sequential recommendation performance. For example, studies on hybrid optimization models for real-world applications [35], AI-driven optimization frameworks [36], and advanced evolutionary computation methods [37] provide valuable insights into integrating optimization strategies with deep learning architectures. Incorporating these techniques can lead to improved trade-offs between accuracy, diversity, and fairness, thereby further validating the contributions of our study.

Contrastive learning for recommendation

The application of contrastive learning in sequential recommendation is a recent development aimed at improving model robustness and representation quality. Previous work has applied contrastive learning strategies to the field of sequential recommendation, focusing on enhancing the model's representation capabilities. In these studies, data augmentation is mainly categorized into two types: sequence augmentation [3, 9, 11, 38, 39] and target item augmentation [40–43]. Common methods for sequence augmentation include masking, reordering, cropping, replacement, and insertion, while target item augmentation often employs dropout methods and other related techniques.

Beyond conventional collaborative filtering (CF) techniques, recent studies have explored topic modeling and textual analysis to improve recommendation quality. For instance, some works employ Latent Dirichlet Allocation (LDA) to extract latent associations between items and user preferences, thereby enhancing personalization [44]. These methods help address limitations of traditional CF models by leveraging additional content features for improved recommendation quality.

Despite the continuous emergence of data augmentation methods, these methods generally overlook the issues of insufficient contrastive learning and model training dominated by popular items caused by the popularity bias, which if not properly addressed, will directly affect the effectiveness of contrastive learning. To this end, this study proposes a novel asymmetric multi-instance data augmentation strategy, which generates different numbers of augmented views for popular and unpopular items, and uses the

proposed AMINCE loss function for contrastive learning to effectively alleviate the aforementioned problems.

Background

Single-Instance augmentation. The traditional Noise Contrastive Estimation (NCE) [45] primarily facilitates the comparison of representations in the latent space, encouraging the predicted representation $\widehat{\mathbf{z}}$ to be close to the target representation $\mathbf{z}$. The predicted representation $\widehat{\mathbf{z}}$ is typically derived from the embedding of augmented data. During the forward pass, previous works have computed the latent vectors for both the predicted representation $\widehat{\mathbf{z}}$ and the target representation $\mathbf{z}$. The similarity between these latent vectors is measured using the dot product $\widehat{\mathbf{z}}_i^\top \cdot \mathbf{z}_i$. After generating a semantic positive sample for each item in the batch, the NCE loss incorporates a single term in the numerator, while the denominator retains its original form:

$$\ell_{\mathrm{NCE}} = -\frac{1}{B}\sum_{i=1}^{B}\left[\log \frac{e^{(\widehat{\mathbf{z}}_i^\top \cdot \mathbf{z}_i)/\tau}}{e^{(\widehat{\mathbf{z}}_i^\top \cdot \mathbf{z}_i)/\tau} + \sum_{j\in\mathcal{N}_i} e^{(\widehat{\mathbf{z}}_i^\top \cdot \mathbf{z}_j)/\tau}}\right]. \tag{1}$$

Here, τ denotes the temperature coefficient, and $\mathcal{N}_i$ represents the negative sample set for item i. Essentially, Equation (1) equates to the cross-entropy loss between the positive pair and all other negative pairs. For the predicted vector $\widehat{\mathbf{z}}_i$, the sole positive pair is $(\widehat{\mathbf{z}}_i^\top \cdot \mathbf{z}_i)$, where $\mathbf{z}_i$ is the corresponding target item embedding. All other latent representations $\mathbf{z}_j$ from any other item constitute the negative pairs.

TCA4Rec

Problem definition

In the domain of sequential recommendation, the main task involves modeling user-item interactions as a sequence, which necessitates a structured approach to capture user preferences through time. We define a set of users $\mathcal{U}$ and a set of items $\mathcal{I}$, with each user $u \in \mathcal{U}$ and item $i \in \mathcal{I}$ being unique entities. $|\mathcal{U}|$ and $|\mathcal{I}|$ represent the number of users and items, respectively. For a given user u, the interactions are sequenced in a historical order S_u, denoted as $[i_1, i_2, \ldots, i_n]$, where n is the count of interactions. Alongside the items, we have a comprehensive attribute set $\mathcal{A}$ that includes all attributes across the item set. The total number of distinct attributes is denoted by $|\mathcal{A}|$. Each item i is associated with a subset of attributes $\mathcal{A}_i = \{a_1, a_2, \ldots, a_m\}$, with m being the number of attributes for that item. It is noteworthy that items of different types may exhibit different attribute sets.

The goal of a sequential recommendation system is to predict the next item i_{t+1} in the sequence that user u is likely to interact with, based on their prior interaction sequence $[i_1, i_2, \ldots, i_t]$ and the attributes $\mathcal{A}_i$ of each item. This prediction is made at a specific time step t, aiming to enhance the relevance and accuracy of recommendations.

This framework allows for a nuanced understanding of user preferences by considering both the sequence of interactions and the attributes of items, providing a foundation for developing sophisticated recommendation algorithms.

Contrastive learning framework

The overview of TCA4Rec is illustrated in Fig. 2. Given a sequence of items, each item's ID and attributes are first transformed into dense feature vectors via two embedding layers, Emb_I and Emb_A, respectively. These feature vectors are then processed by an attribute encoder, g_{enc}, which generates a latent representation $\mathbf{z}$ that captures local item-specific features. To incorporate global sequential context, a temporal aggregation module, g_{ta}, encodes the sequential history up to time step t, producing a contextual representation $\mathbf{c}_t$. Specifically, for each item i_t in the sequence, its ID and attributes are encoded into a latent vector $\mathbf{z}_t$ via g_{enc}. The temporal aggregation module, which can be implemented using architectures such as GRU or Transformer, processes the historical sequence to produce $\mathbf{c}_t$, thereby encapsulating global contextual information from past interactions. During the prediction step at $t+1$, the memory function g_m leverages $\mathbf{c}_t$ to generate a predicted latent representation $\hat{\mathbf{z}}_{t+1}$. This predicted representation is then compared to the ground truth latent vector $\mathbf{z}_{t+1}$, derived from item i_{t+1} and its attributes. The similarity between $\hat{\mathbf{z}}_{t+1}$ and $\mathbf{z}_{t+1}$ is measured using the AMINCE loss, as depicted in Fig. 2 for a one-step prediction.

To facilitate contrastive learning, different types of samples are drawn from the training mini-batch: (1) positive samples (Pos.), (2) semantic positive samples (Sem. Pos.), and (3) general negative samples (General Neg.). Specifically, the semantic positive samples, $\mathbf{z}^1_{t+1}, \ldots, \mathbf{z}^*_{t+1}$, are obtained by applying different Dropout variations on the same g_{enc}. The general negative samples, $\mathbf{z}^{'}_{t+*}$, are selected from other sequences within the same batch. The AMINCE loss strategically omits the Pos. samples in the gray box. Notably, if Pos. samples are used instead of Sem. Pos., AMINCE simplifies to the standard NCE loss. Moreover, the number of Sem. Pos. is different, which is decided by $\mathbf{z}_{t+1}$ is popular or nonpopular item.

During inference, recommendation results are obtained by comparing the similarity between the latent representations $\mathbf{z}$ of candidate items and the encoded user preference representation. This process ensures that the model effectively captures both short-term and long-term sequential dependencies for robust recommendation performance.

1) Embedding layer: TAC4Rec utilizes two embedding matrices to covert item and attribute identifiers into a dense vector space, where $\text{Emb}_I \in \mathbb{R}^{|I|\times d}$ is item embedding matrix and $\text{Emb}_A \in \mathbb{R}^{|A|\times d}$ is the attribute embedding matrix. d denotes the

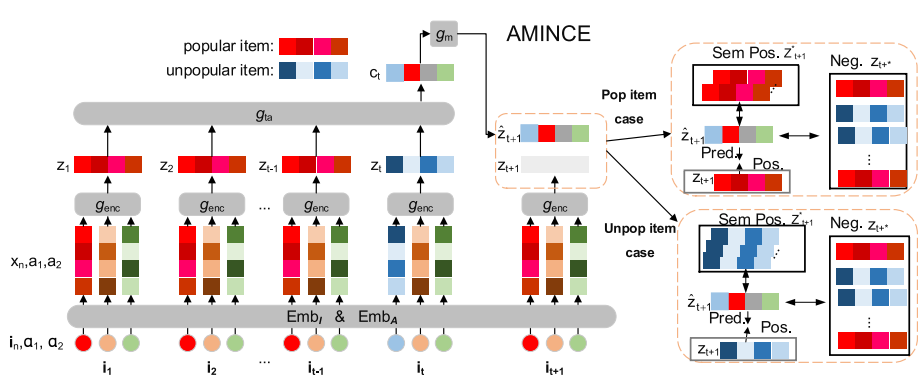

Fig. 2 Overview of the TCA4Rec architecture

embedding size. For example, given an item i_n with its attributes $\{a_1, a_2\}$, the corresponding dense vectors are calculated as follows:

$$\mathbf{x}_n = \mathrm{Emb}_I(i_n), \tag{2}$$

$$\mathbf{a}_k = \mathrm{Emb}_A(a_k), k \in \{1, 2\}. \tag{3}$$

2) Feature integration module: The feature fusion aims to fuse all the information of the item, including the ID information and the side information, into a latent representation. Specifically, when it obtain the item embedding $\mathbf{x}$ and its attribute embeddings $\mathbf{a}_*$ an feature fusion function The feature fusion layer consolidates the item's identifier and its associated attributes into a unified latent representation, leveraging the richness of item features to enhance recommendation accuracy:

$$\mathbf{z} = \mathrm{g}_{\mathrm{enc}}(\mathbf{x}, \{\mathbf{a}_*\}), \tag{4}$$

where $\mathbf{x}$, $\mathbf{a}_*$, and $\mathbf{z}$ represent the item, attribute, and latent representations, respectively.

3) Temporal aggregation module: This module is designed to aggregates temporal dynamic of items across the sequence until current time t. It is defined as g_{ta}:

$$\mathbf{c}_t = \mathrm{g}_{\mathrm{ta}}(\mathbf{z}_1, \mathbf{z}_2, \ldots, \mathbf{z}_t), \tag{5}$$

where the output $\mathbf{c}_t \in \mathbb{R}^{1\times d}$. g_{ta} can be implement by GRU or Attention models.

4) Memory argumentation module: To enhance the model's predictive capabilities, the memory module employs a memory bank ($\mathbf{M} \in \mathbb{R}^{b\times d}$ with b memory slots) to generate outputs based on the context vector. It is defined as g_m:

$$\widehat{\mathbf{z}}_{t+1} = \mathrm{g}_m(\mathbf{c}_t) = \mathrm{Softmax}(\mathrm{MLP}(\mathbf{c}_t)) \cdot \mathbf{M} + \mathbf{c}_t, \tag{6}$$

where MLP denotes a multi-layer perceptron, and residual connections are used to maintain prediction integrity.

5) Recommendation engine: In real-world scenarios, the model scores the sequence representation against all items, ranking them based on the dot product between the sequence representation and item latent representations.

Two-stage training strategy with TAMINCE

Noise Contrastive Estimation (NCE) [45] is a classification objective designed to distinguish real samples from noisy ones. Following the approach in [46], NCE can be directly applied to the item prediction task. In TCA4Rec, we extend NCE to an asymmetric multi-instance variant integrated with a two-stage training strategy-termed *Two-stage Asymmetric Multi-instance Noise Contrastive Estimation (TAMINCE)*-which effectively alleviates both sparse training signals and popularity bias. The TAMINCE framework operates in two distinct stages:

Stage 1: Contrastive Representation Learning with AMINCE. In Stage 1, an asymmetric multi-instance augmentation strategy (AMINCE) is employed to learn robust item representations across the dataset. This stage enriches the training signal by generating multiple augmented views for each item via the following components:

(1) *Batch-wise Negative Sampling:* The contrastive loss is computed within each mini-batch rather than over the entire item set, as a mini-batch typically provides a sufficient number of samples [47]. Drawing inspiration from Contrastive Predictive Coding (CPC) [48] in computer vision-where negative samples are selected from different channels of the same feature map or from other feature maps within the same batch-we construct the negative sample set $\mathcal{N}_i$ for each item i using the feature vectors $\mathbf{z}$ of all other items in the mini-batch. Items from different sequences in the mini-batch naturally serve as negative samples, as they likely reflect distinct user preferences. Consequently, these items are treated as "easy negatives" in (7) and (8). As illustrated in Fig. 2 (labeled as *Neg* in the AMINCE loss), if there are D unique items in the batch, then $|\mathcal{N}_i| = D - 1$.

(2) *Popularity-aware Asymmetric Positive Sampling:* For a predicted latent representation $\hat{\mathbf{z}}_i$, the most straightforward positive pair is $(\hat{\mathbf{z}}_i, \mathbf{z}_i)$, where $\mathbf{z}_i$ is the ground-truth latent vector at the corresponding time step. To address sparse training signals and popularity bias, we extend this strategy by generating multiple semantically positive samples. Identifying semantically similar positive samples for next-item prediction is challenging since the prediction target is the next item, and sequence-level semantic similarity is hard to define. Moreover, in long-tail distributions, popular items appear frequently while unpopular items are less common. Therefore, we generate fewer augmented views (i.e., k views) for popular items and more augmented views (i.e., $k + n$ views) for unpopular items. This asymmetric augmentation ensures that, despite their low frequency, unpopular items are sufficiently represented during contrastive learning, thereby enabling the model to learn more discriminative item representations. To implement this, we partition the item set $\mathcal{I}$ into two subsets based on popularity with a threshold r_{tails}: the popular item subset $\mathcal{I}^p$ and the long-trails item subset $\mathcal{I}^u$. As depicted in Fig. 2, an item i is encoded into a dense embedding vector $\mathbf{z}_i$ via the encoder g_{enc}. With Dropout applied within g_{enc}, multiple semantically similar yet distinct latent vectors are generated by varying the Dropout masks. Specifically, for popular items, k distinct Dropout masks yield the set $\mathcal{P}_i = \{\mathbf{z}_i^1, \mathbf{z}_i^2, \ldots, \mathbf{z}_i^k\}$; for unpopular items, $k + n$ masks produce the set $\overline{\mathcal{P}}_i = \{\mathbf{z}_i^1, \mathbf{z}_i^2, \ldots, \mathbf{z}_i^{k+n}\}$.

(3) *Multi-instance Contrastive Loss:* Utilizing the semantic positive sets and the negative sample set, the multi-instance contrastive loss (AMINCE) is defined as:

$$\ell_{\text{AMINCE}_p} = \log \frac{\sum_{j \in \mathcal{P}_i} e^{(\widehat{\mathbf{z}}_i^\top \mathbf{z}_j)/\tau}}{\sum_{j \in \mathcal{P}_i} e^{(\widehat{\mathbf{z}}_i^\top \mathbf{z}_j)/\tau} + \sum_{j \in \mathcal{N}_i} e^{(\widehat{\mathbf{z}}_i^\top \mathbf{z}_j)/\tau}}, \tag{7}$$

$$\ell_{\text{AMINCE}_{\overline{p}}} = \log \frac{\sum_{j \in \overline{\mathcal{P}}_i} e^{(\widehat{\mathbf{z}}_i^\top \mathbf{z}_j)/\tau}}{\sum_{j \in \overline{\mathcal{P}}_i} e^{(\widehat{\mathbf{z}}_i^\top \mathbf{z}_j)/\tau} + \sum_{j \in \mathcal{N}_i} e^{(\widehat{\mathbf{z}}_i^\top \mathbf{z}_j)/\tau}}, \tag{8}$$

Here, $\mathcal{N}_i$ is the negative sample set for item i, containing k variants for each negative sample drawn from the batch, so that $|\mathcal{N}_i| = k(D-1)$, τ is a temperature parameter.

Stage 2: Debiasing via selective update strategy. In Stage 2, we introduce a selective update strategy to mitigate the over-dominance of popular items during training. In each training iteration, only a random subset of popular items is updated, while the remaining popular items are masked. In contrast, long-tail (unpopular) items are always updated, ensuring that their representations are consistently refined. This selective update strategy rebalances the contrastive learning process by enabling a more equitable comparison between popular and non-popular items.

Integrating the components from both stages, the overall TAMINCE framework is formalized as follows:

$$\ell_{\text{TAMINCE}_{S1}} = -\frac{\sum_{i \in \mathcal{I}_p} \ell_{\text{AMINCE}_p} + \sum_{i \in \mathcal{I}_{\overline{p}}} \ell_{\text{AMINCE}_{\overline{p}}}}{|\mathcal{I}_p| + |\mathcal{I}_{\overline{p}}|}, \tag{9}$$

$$\ell_{\text{TAMINCE}_{S2}} = -\frac{\sum_{i \in \widetilde{\mathcal{I}}_p} \ell_{\text{AMINCE}_p} + \sum_{i \in \mathcal{I}_{\overline{p}}} \ell_{\text{AMINCE}_{\overline{p}}}}{\left|\widetilde{\mathcal{I}}_p\right| + |\mathcal{I}_{\overline{p}}|}, \tag{10}$$

$$\widetilde{\mathcal{I}}_p = \text{Mask}(\mathcal{I}_p), \tag{11}$$

where $\widetilde{\mathcal{I}}_p$ denotes the subset of popular items remaining after random masking, $\mathcal{I}_p$ represents the complete set of popular items, and $\mathcal{I}_{\overline{p}}$ (or equivalently $\mathcal{I}^u$) denotes the set of unpopular items.

Algorithm 1 summarizes the training process of TCA4Rec. The pseudocode details the main steps including embedding generation, feature integration, temporal aggregation, memory augmentation, contrastive loss computation for both popular and long-tail items, and parameter updates across two training stages.

Algorithm 1 Learning Algorithm of TCA4Rec

Initialize the maximum number of training epochs T_e;
Initialize the batch size B;
Partition the item set $\mathcal{I}$ into two subsets based on the popularity threshold r_{tails}: popular items $\mathcal{I}^p$ and long-tail items $\mathcal{I}^u$;
for $epoch = 1$ to T_e **do**
  for each minibatch $\{s_u\}_{u=1}^{B}$ **do**
    Compute item embeddings $\{x_1, \ldots, x_n\}$ and attribute embeddings $\{a_1, \ldots, a_k\}$ via the embedding layer with (2);
    Compute fused representations $\{z_1, \ldots, z_n\}$ via the feature integration module with (4);
    Compute temporal aggregated context $\mathbf{c_t}$ via the temporal aggregation module with (5);
    Predict next-step representation $\hat{\mathbf{z}}_{t+1}$ via the memory augmentation module with (6);
    if $epoch \in S1$ **then**
      if $s_u^{t+1} \in \mathcal{I}^p$ **then**
        Obtain k semantic-positive samples $(z_{t+1}^1, \ldots, z_{t+1}^k)$ and $|\mathcal{N}_i|$ negative samples $(z'_1, \ldots, z'_{|\mathcal{N}_i|})$ via the fusion module with *Dropout*;
        Compute the contrastive loss $\ell_{\text{AMINCE}_{s_u^{t+1}}}$ (7);
      else if $s_u^{t+1} \in \mathcal{I}^u$ **then**
        Obtain $k + n$ semantic-positive samples $(z_{t+1}^1, \ldots, z_{t+1}^{k+n})$ and $|\mathcal{N}_i|$ negative samples $(z'_1, \ldots, z'_{|\mathcal{N}_i|})$ via the fusion module with *Dropout*;
        Compute the contrastive loss $\ell_{\text{AMINCE}_{s_u^{t+1}}}$ (8);
      end if
      Aggregate and compute the overall loss $\ell_{\text{TAMINCE}_{S1}}$ (9);
      Update TCA4Rec parameters by minimizing $\ell_{\text{TAMINCE}_{S1}}$;
    end if
    if $epoch \in S2$ **then**
      Compute masked popular item set $\widetilde{\mathcal{I}}^p$ (11);
      Perform asymmetric data augmentation on the remaining items via the fusion module with *Dropout*;
      Compute the contrastive loss $\ell_{\text{TAMINCE}_{S2}}$ (10);
      Update TCA4Rec parameters by minimizing $\ell_{\text{TAMINCE}_{S2}}$;
    end if
  end for
end for

Discussion

In this section, we discuss several key aspects of our proposed TCA4Rec framework.

Comparison with traditional contrastive losses: Traditional contrastive learning approaches, such as the Noise Contrastive Estimation (NCE) objective, typically utilize a single ground truth positive sample along with multiple negative samples per instance. Although this method is sample-efficient, it provides a limited supervision signal and struggles to address the sparse interactions common in sequential recommendation tasks. The Multi-instance Noise Contrastive Estimation (MINCE) loss partially alleviates this issue by incorporating multiple semantically positive and negative samples. However, both NCE and MINCE fall short in mitigating the inherent popularity bias, where a small subset of popular items dominates user interactions. In contrast, the AMINCE

loss used in TCA4Rec introduces two key innovations: (1) an asymmetric augmentation strategy that generates fewer augmented views for popular items and more views for non-popular (long-tail) items, thereby enriching the supervision signal for underrepresented items, and (2) a two-stage training strategy. In Stage 1, robust representations are learned using the asymmetric multi-instance approach, while in Stage 2, a selective update mechanism ensures that only a random subset of popular items is updated, with non-popular items continuously refined. This dual strategy re-balances the learning process and effectively mitigates popularity bias.

Multi-modal extensions of TCA4Rec framework: The design of TCA4Rec naturally accommodates the integration of multi-modal signals. TCA4Rec's clear separation into an attribute encoder, temporal aggregation module and memory function. The attribute encoder can be extended to process diverse side information (e.g., text, image, and structured metadata) using modality-specific encoders or pre-trained vision-language models. For example, one can replace the attribute encoder with a multi-modal encoder that concatenates text embeddings (from a transformer) and image embeddings (from a CNN) before projecting into $\mathbf{z}_t$, or augment the memory function with a graph-based module to capture item-item relations. This opens up opportunities to enhance recommendation robustness and personalization by leveraging richer semantic cues. Recent advances in multi-modal contrastive learning and adaptive data augmentation further provide plug-and-play components that can be readily integrated into TCA4Rec's architecture.

Experiments

In this section, we conduct extensive experiments to validate the effectiveness of TCA4Rec, where we mainly focus on the following research questions (RQ):

- **RQ1:** How does the performance of TCA4Rec compare to that of state-of-the-art sequential recommendation models in terms of overall recommendation accuracy, running time and accuracy on long-term user interactions?
- **RQ2:** Whether our framework can maintain the recommendation performance in face of the different degree of the sparsity data and random noise ?
- **RQ3:** How does the key component in our framework contribute to the final performance?
- **RQ4:** How do the number of augmented views n and popularity thresholds influence the performance of TCA4Rec?

In the following, we firstly introduce the experiment setup, and then present and analyze the results to answer the above questions.

Experiment setup

Datasets. To evaluate the proposed TCA4Rec model, we conduct our experiments on the following three real-world datasets: Beauty, Sports, and Toys [49],[1] which are all

[1] http://jmcauley.ucsd.edu/data/amazon/

Table 1 Statistics of the experimental datasets

Dataset	Beauty	Sports	Toys
Users	22,363	35,598	19,412
Items	12,101	18,357	11,924
Actions	198,502	296,337	167,597
Attributes	1221	2277	1027
Avg. Actions / User	8.9	8.3	8.6
Avg. Actions / Item	16.4	16.1	14.1
Avg. Attribute / Item	5.1	6.0	4.3
Sparsity	99.93%	99.95%	99.93%

collected from Amazon and contain the user purchasing records on beauty, sports and toys products, respectively. The dataset statistics are summarized in Table 1.

Prepossessing. For all datasets, we treat each user's interaction records as a sequence, where the order is determined by the timestamp. Following [7, 39, 50], we filter all datasets that all users and items have at least 5 interactions. The maximum length of a sequence is limited to 50. That is if a sequences longer than 50 interactions, only the 50 most recent interactions are retained. Following [11, 46, 51], we adopt a *leave-one-out* evaluation protocol, which is commonly used for sequential recommendation models. Specifically, for each user u, the last item in the sequence is reserved for testing, the second-to-last item is used for validation, and all preceding items constitute the training set.

Baselines

To demonstrate the effectiveness and generality of TCA4Rec, the following representative sequential recommendation models are chosen for comparison:

- **GRU4Rec** [13]**:** This model leverages the power of Gated Recurrent Units (GRU) to encapsulate the sequential nature of user interactions. By treating each sequence as an input, GRU4Rec adeptly captures the evolving preferences of users through the hidden states produced at each timestep, offering robust support for sequential recommendation tasks.
- **Caser** [8]**:** By utilizing Convolutional Neural Networks (CNNs), this approach employs convolutions in both horizontal and vertical dimensions to extract features indicative of high-order Markov Chains. This technique sets it apart from conventional sequence modeling architectures by providing a unique perspective on sequential data.
- **SASRec** [14]**:** SASRec stands out as a unidirectional, self-attention-based model designed for sequential recommendations. Its efficacy lies in the multi-head attention mechanism that intelligently selects the most relevant items from the sequence, thus enhancing its predictive capabilities for next-item recommendations.
- **CL4SRec** [11]**:** Pioneering the integration of contrastive learning within the realm of sequential recommendations, CL4SRec harnesses data augmentation to create positive sample pairs. This model particularly emphasizes the use of random augmenta-

tions to generate a rich set of training signals, enhancing the representation learning process.

- **S^3Rec** [39]**:** This model introduces masked contrastive pre-training to the field of sequential recommendation, focusing on segment, attribute, and item-level masking. It serves as a formidable baseline, particularly in scenarios where side information about items is available to bolster recommendation performance.
- **ICLRec** [52]**:** Focusing on clustering to distill user interests, ICLRec represents user intentions through cluster centroids. It formulates a composite loss function that integrates user intention with interest representation, aligning with both the sequence loss and contrastive learning objectives to achieve multi-task optimization.
- **MMInfoRec** [46]**:** MMInfoRec presents a cutting-edge framework that combines memory-augmented techniques with multi-instance contrastive predictive coding. This approach streamlines the training process by eliminating the need for fine-tuning, allowing for a more efficient path to optimal recommendation performance.
- **BERT4Rec** [7]: it is a transformer-based sequential recommendation model that employs a masked item training scheme using a deep bidirectional self-attention mechanism. This approach prevents information leakage by adopting a Cloze objective, where random masked items are predicted based on both left and right context, thus enabling the model to learn comprehensive bidirectional representations.
- **TiSASRec** [53]: it is a time interval-aware self-attention framework for sequential recommendation. In TiSASRec, both the absolute positions of items and the time intervals between interactions are explicitly modeled within a transformer-based architecture, capturing the influence of varying time intervals on the next-item prediction.

Evaluation protocols

In our experiments, we employ Hit Ratio (HR@K) and Normalized Discounted Cumulative Gain (NDCG@K) to measure recommendation performance, which are two widely used metrics of Top-K recommendation [3, 7, 39, 46]. For each dataset, $|\mathcal{U}|$ and $|\mathcal{I}|$ denote the size of user and item sets, receptively. $\mathcal{L}_u$ is a list with K recommendation items generated by sequential recommender, while Γ_u denotes the item set rated by user u in the test set.

HR@K: it is a widely used metric to measure the recommendation accuracy of a sequential recommender, it can be expressed as:

$$\text{HR}@K = \frac{1}{|\mathcal{U}|} \times \sum_{u \in \mathcal{U}} \frac{\text{hits}(u)}{K}, \tag{12}$$

For user u, hits(u) is the interaction between the K recommendations generates by sequential recommender and test items for u (i.e., $\text{hits}(u) = |\mathcal{L}_u \cap \Gamma_u|$).

NDCG@K: it concerns the hit's position via calculating a gain for each item in the recommendation list, which assigns higher scores to hits at top ranks:

$$NDCG@K = Z_K \sum_{i=1}^{K} \frac{2^{r_i} - 1}{\log_2(i+1)}, \tag{13}$$

where Z_K is the normalizer to ensure the value of $NDCG@K$ in the range of [0,1]. r_i is the gain from being recommended an item at position i in the list. For simplicity, we set $r_i = 1$ if the recommended item is in the test set, otherwise $r_i = 0$.

We also adopt the cross-validation to evaluate the recommender performance. By default, we set K = 5,10 for each dataset. All experiments are run 5 times, and the average and standard deviation are reported.

Parameter settings. To ensure a fair comparison, the hyperparameters for all methods were initialized according to their original literature and subsequently fine-tuned on the validation set using grid search. For TCA4Rec, the embedding size is fixed at 64 and is consistently applied across all mapping operations. The number of layers and heads in the Transformer are selected from the set $\{1, 2\}$. To prevent over-fitting, a Dropout function with a ratio of 0.5 is applied both to the input of the temporal aggregation function g_{ta} and within the Transformer module inside g_{ta}. The training batch size is set to 256, and the Adam optimizer is used with the learning rate chosen from $\{0.0001, 0.0005, 0.001, 0.005, 0.01, 0.05\}$. The number of memory slots b is selected from $\{5, 10, 32, 64, 128, 256\}$, with the default prediction step set to 1. The temperature parameter τ in (7) and (8) is chosen from $\{0.1, 0.3, 0.6, 1, 3\}$. L2 regularization is applied during training with weights selected from $\{0, 0.1, 0.01, 0.001, 0.0001, 0.00001\}$. Finally, the number of augmented views n for non-popular items is optimized over the range $[0, 1, 2, \ldots, 10]$.

Overall performance (RQ1)

Accuracy comparison. We first conduct a comparative analysis between TCA4Rec and state-of-the-art methods with different evaluation metrics. Table 2 shows the performance comparison on Amazon Beauty, Sports and Toys. From Table 2, we can have the following observations:

- TCA4Rec demonstrates a consistent and significant advantage across all datasets and metrics. It notably surpasses the nearest competitor on the Beauty dataset by 15.71% in HR@5 and 14.10% in NDCG@5, with relative improvements of 13.15% in HR@10 and 13.35% in NDCG@10. These gains are echoed on the Sports and Toys datasets, with TCA4Rec achieving the highest relative improvements of 19.26% in HR@5 and 17.95% in NDCG@5 on Sports, and 12.20% in HR@5 and 10.57% in NDCG@5 on Toys. These enhancements validate the robustness and effectiveness of TCA4Rec in handling the challenges of sparse supervision signals and popularity bias in training data.
- The performance gap between traditional methods like GRU4Rec and Caser, which rely on basic RNN and CNN architectures, and more advanced models is evident. These methods tend to produce less competitive results due to their limited capability to capture complex sequential patterns. In contrast, SASRec, which employs attention mechanisms, significantly outperforms these traditional approaches, underscoring the importance of leveraging Transformer-based architectures for capturing intricate item correlations within sequences.
- Models employing contrastive learning, such as CL4SRec, S^3Rec, ICLRec, and MMInfoRec, show varying degrees of improvement over traditional baselines.

Table 2 Recommendation performance of all methods on the three datasets

Dataset	Methods	Top-5		Top-10	
		HR@5↑	NDCG@5↑	HR@10↑	NDCG@10↑
Beauty	GRU4Rec	0.0164	0.0099	0.0283	0.0137
	Caser	0.0205	0.0131	0.0347	0.0176
	SASRec	0.0387	0.0249	0.0605	0.0318
	CL4SRec	0.0396	0.0232	0.0630	0.0307
	S^3Rec	0.0387	0.0244	0.0647	0.0327
	ICLRec	0.0492	0.0327	0.0716	0.0400
	MMInfoRec	0.0541	0.0390	0.0753	0.0457
	TiSASRec	0.0601	0.0337	0.0793	0.0426
	BERT4Rec	0.0180	0.0110	0.0322	0.0156
	TCA4Rec	**0.0626**	**0.0445**	**0.0852**	**0.0518**
	%improv.	4.16%	14.10%	7.44%	13.35%
Sports	GRU4Rec	0.0129	0.0086	0.0204	0.0110
	Caser	0.0116	0.0072	0.0194	0.0097
	SASRec	0.0233	0.0154	0.0350	0.0192
	CL4SRec	0.0256	0.0164	0.0374	0.0207
	S^3Rec	0.0251	0.0161	0.0385	0.0204
	ICLRec	0.0256	0.0168	0.0386	0.0215
	MMInfoRec	0.0270	0.0195	0.0387	0.0233
	TiSASRec	0.0240	0.0143	0.0381	0.0188
	BERT4Rec	0.0081	0.0050	0.0146	0.0156
	TCA4Rec	**0.0322**	**0.0230**	**0.0431**	**0.0265**
	%improv.	19.26%	17.95%	11.37%	13.73%
Toys	GRU4Rec	0.0097	0.0059	0.0176	0.0084
	Caser	0.0116	0.0107	0.0270	0.0141
	SASRec	0.0463	0.0306	0.0675	0.0374
	CL4SRec	0.0503	0.0264	0.0755	0.0339
	S^3Rec	0.0443	0.0294	0.0700	0.0376
	ICLRec	0.0543	0.0372	0.0780	0.0449
	MMInfoRec	0.0623	0.0454	0.0815	0.0516
	TiSASRec	0.0601	0.0337	0.0876	0.0426
	BERT4Rec	0.0114	0.0070	0.0219	0.0104
	TCA4Rec	**0.0699**	**0.0502**	**0.0896**	**0.0566**
	%improv.	12.20%	10.57%	2.28%	9.69%

We highlight the best results in bold and sub-optimal results with underline. *%improv.* represents the relative improvement of TCA4Rec over the sub-optimal baseline

CL4SRec introduces contrastive learning to sequential recommendation but with limited performance gains due to its heuristic data augmentation methods. S^3Rec, while integrating self-attention and multiple self-supervised objectives, does not show substantial improvements over SASRec, suggesting misalignment between its contrastive tasks and the recommendation goal. ICLRec, focusing on modeling user intent, shows significant gains, indicating the value of intent-aware contrastive learning. MMInfoRec, with its multi-instance contrastive framework and memory enhancement, becomes a strong contrastive baseline, highlighting the effectiveness of aligning contrastive tasks with recommendation objectives.

- TCA4Rec's superiority over MMInfoRec is attributed to its tailored augmentation strategies for items of varying popularity, which bolsters the model's representational

Table 3 Recommendation performance of different models for users with long-term interactions

Method	Beauty		Sports		Toys	
	HR@5	NDCG@5	HR@5	NDCG@5	HR@5	NDCG@5
TiSASRec	0.0229	0.0136	0.0601	0.0357	0.0533	0.0298
MMInfoRec	0.0253	0.0179	0.0647	0.0466	0.0532	0.0379
TCA4Rec	**0.0270**	**0.0192**	**0.0718**	**0.0510**	**0.0558**	**0.0393**

Table 4 Running time comparison of each model (Recommendation and Training Epoch)

Method	Beauty		Sports		Toys	
	Rec. (s)	Training (s)	Rec. (s)	Training (s)	Rec. (s)	Training (s)
BERT4Rec	2.8	42	4.5	63	2.5	35
TiSASRec	2.3	78	3.8	112	2.2	64
MMInfoRec	6.1	68	12.1	99	5.4	61
TCA4Rec	6.3	91	12.8	124	5.6	68

capabilities and recommendation performance. This comparison underscores the importance of asymmetric augmentation in addressing the long-tail challenge prevalent in recommendation systems.

- To further contextualize the performance of TCA4Rec, we extend our comparison to include two representative transformer-based sequential recommendation models: BERT4Rec and TiSASRec. BERT4Rec leverages a masked item training scheme with deep bidirectional self-attention, predicting randomly masked items based on both left and right contexts. TiSASRec, in contrast, integrates time interval awareness into its transformer-based framework, modeling absolute positions and interaction intervals to enhance sequential modeling. As shown in Table 2, TCA4Rec consistently outperforms these models, particularly in mitigating popularity bias and improving the representation of long-tail items.

Accuracy on long-term user interactions. To assess the effectiveness of TCA4Rec in capturing users' evolving interests over time, we evaluated the recommendation performance on users with long-term interactions (i.e., the top one-third of users based on interaction length). Table 3 presents the performance comparison across three datasets. As shown in Table 3, TCA4Rec outperforms both TiSASRec and MMInfoRec across all datasets. For example, on the Beauty dataset, TCA4Rec achieves an HR@5 of 0.0270 and an NDCG@5 of 0.0192, compared to 0.0229 and 0.0136 for TiSASRec, and 0.0253 and 0.0179 for MMInfoRec, respectively. Similar performance improvements are observed on the Sports and Toys datasets. These results indicate that TCA4Rec is particularly effective in modeling long-term user interactions and capturing evolving user preferences, thus ensuring improved recommendation accuracy for users with extensive interaction histories.

Computational time comparison. Table 4 presents a comprehensive comparison of the running time per valid epoch and per training epoch for each model across

the Beauty, Sports, and Toys datasets. Although TCA4Rec requires a modest increase in training time relative to MMInfoRec (approximately 34%, 25%, and 11% longer on Beauty, Sports, and Toys, respectively), its validation time is comparable to that of the baselines. These additional computational costs are offset by significant improvements in recommendation accuracy-such as a 14.1% improvement in NDCG@5 on Beauty, 19.26% in HR@5 on Sports, and 12.2% in HR@5 on Toys-demonstrating a favorable trade-off between performance and efficiency. Importantly, the inference time of TCA4Rec remains similar to that of the baseline methods, ensuring its viability for real-time recommendation pipelines.

Robustness investigation (RQ2)

To comprehensively assess the robustness of TCA4Rec, we conduct a series of experiments to evaluate its performance under varying degrees of data sparsity and random noise.

Impact of data sparsity. To examine the effect of data sparsity, we maintain a fixed testing set while constructing multiple training subsets by varying the proportion of available items (25%, 50%, 75%, and 100%). We compare TCA4Rec against SASRec and two competitive baselines, MMInfoRec and ICL4Rec. The results across three datasets are illustrated in Fig. 3.

As shown in Fig. 3, the recommendation accuracy (measured by HR@5 and NDCG@5) of all models significantly declines as the amount of training data decreases. However, TCA4Rec consistently exhibits superior robustness compared to the baselines, showing a slower performance degradation. For instance, on the Beauty dataset (HR@5), TCA4Rec trained with only 75% of the data achieves comparable performance to MMInfoRec trained with 100% of the data. When the training data is reduced to 50%, MMInfoRec experiences a 38.81% performance drop, while ICL4Rec declines by 45.93%. In contrast, TCA4Rec exhibits a more moderate decline of 34.38%, demonstrating its resilience against data sparsity. At the extreme case of 25% training data, ICL4Rec suffers

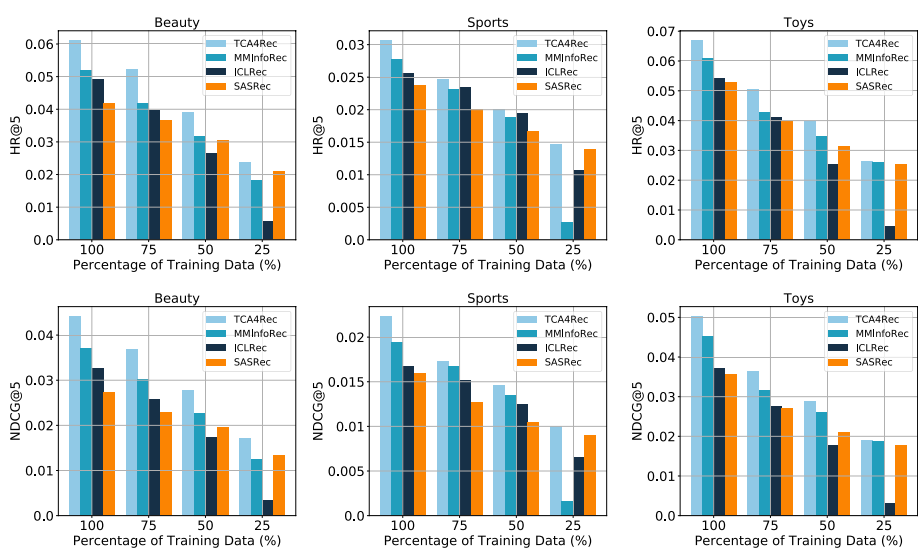

Fig. 3 Performance comparison over different sparsity ratios

severe degradation, while TCA4Rec maintains relatively stable performance. These results suggest that the asymmetric augmentation strategy in TCA4Rec effectively alleviates the impact of sparsity.

Moreover, the effect of sparsity varies across datasets. For example, when using only 50% of training data, TCA4Rec's performance declines by 34.38% on the Beauty dataset, 34.31% on the Sports dataset, and 40.98% on the Toys dataset, indicating dataset-specific sensitivity to sparsity.

Impact of random noise.To assess TCA4Rec's resilience against noisy interactions, we introduce varying levels of random noise (10%, 20%, and 30%) into the training data while keeping the testing set unchanged. The corresponding results are depicted in Fig. 4.

As expected, all models exhibit performance degradation as the noise ratio increases. Noisy interactions mislead recommendation models by introducing incorrect item transition patterns, thereby impacting accuracy. However, TCA4Rec consistently outperforms the baselines, demonstrating a more gradual performance decline as noise levels increase. This highlights its robustness in learning meaningful representations even in the presence of corrupted interactions.

Overall, our experimental results confirm that TCA4Rec maintains superior robustness across different sparsity levels and noise ratios. By leveraging the asymmetric augmentation strategy, our method effectively mitigates the impact of sparse and noisy data, ensuring stable performance across various recommendation scenarios.

Ablation study (RQ3)

We then conduct an ablation study to validate the effectiveness of the key components (i.e., asymmetric multi-instance augmentation and two-stage training strategy) in TCA4Rec. Thus, we define the following variant models: (1) MMInfoRec:it is the base model. When removing asymmetric multi-instance augmentation and two-stage

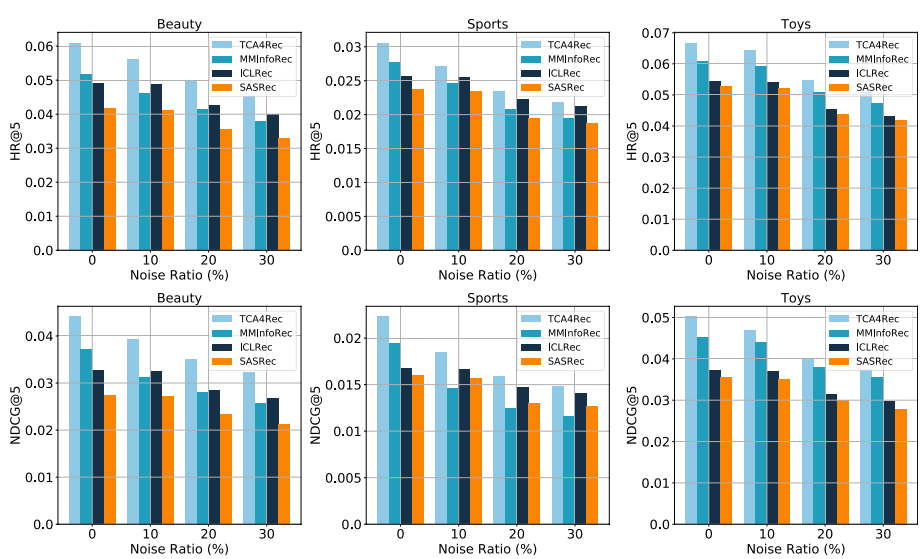

Fig. 4 Performance comparison over different noise ratios

training strategy from TCA4Rec, our model degrades to the standard MMInfoRec model. (2) CA4Rec: in this variant, we measure the contribution of asymmetric multi-instance augmentation function; (3) TC4Rec: this variant focus on evaluating the effectiveness of two-stage training strategy. For each dataset, we sort the items based on popularity. The top 20% designated as popular items, while the rest 80% as non-popular items.

The experimental results on three datasets are shown in Fig. 5. As expected, TCA4Rec outperforms all baseline models on three datasets. Specifically, our base model MMInfoRec already achieves relatively competitive results compared with the mentioned RNN-based and SL-based models, as shown in Table 2. That is because that MMInfoRec utilizes the self-attention mechanism to integrate item IDs with attribute information, as well as aligning contrastive tasks with recommendation tasks to learn strong representations. When the asymmetric multi-instance augmentation module and two-state training module are considered respectively, our variant models (i.e., CA4Rec and TC4Rec) achieve a significant improvement in recommendation accuracy (HR@5). Compared to TC4Rec, the asymmetric multi-instance augmentation module (i.e, CA4Rec) has a prominent impact on accuracy improvement of popularity item subset. Then, incorporating these two modules together, TCA4Rec achieves another significant improvement in recommendation accuracy.

The performance improvement of TCA4Rec is primarily due to the substantial improvement in the popular items. The inner reasons is that the proposed asymmetric multi-instance augmentation module effectively increases the distributional divergence of items in the embedding space, thereby enlarging the distances between items and enhancing the discriminatory of item embedding. This characteristic enables popular items, which have sufficient training samples, to achieve more precise representations during embedding. Thus, our model TCA4Rec can significant improves the accuracy of sequential recommendation via matching popular items with user interests.

HyperParameters analysis (RQ4)

Sensitively of augmented views. In this experimental, we examine the impact of a critical hyperparameter in our framework-the number of augmented views n allocated to non-popular items within the asymmetric multi-instance augmentation

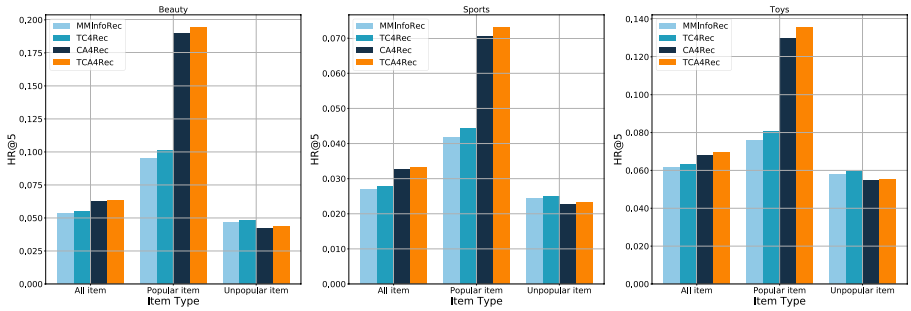

Fig. 5 Ablation experimental results on different datasets

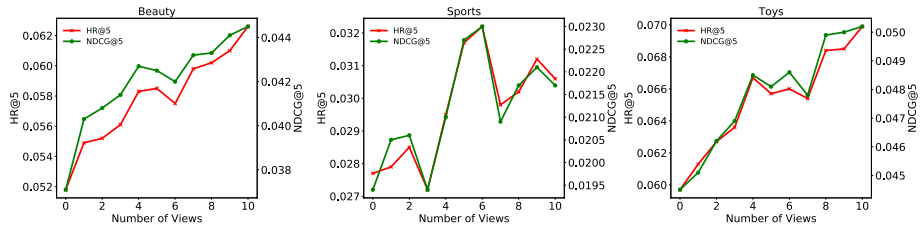

Fig. 6 Model performances for different numbers of augmented views for unpopular items

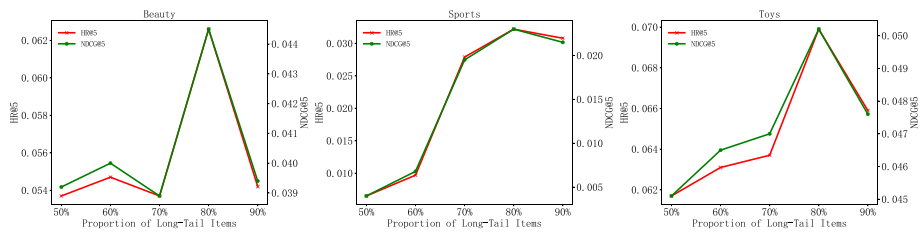

Fig. 7 Performance comparison over different popularity ratios

strategy. This strategy is specifically designed to mitigate popularity bias. We conducted extensive experiments by tuning n across a range from 1 to 10 and evaluated its influence on model performance, as shown in Fig. 6.

The results reveal that even setting $n = 1$ yields a notable performance improvement across all three datasets, underscoring the importance of augmented views in enhancing the representation for less popular items, which typically suffer from underrepresentation in the training data. Furthermore, as n increases, a consistent upward trend in performance is observed, indicating that increasing the number of augmented views for non-popular items effectively counteracts the dominance of popular items during training. The positive correlation between the increment of augmented views n and the overall model performance demonstrates that our augmentation strategy successfully mitigates popularity bias by enriching the representation of non-popular items. On the Beauty and Toys datasets, TCA4Rec exhibits a significant performance surge as n increases, suggesting potential gains beyond $n = 10$. These observations imply that the optimal value of n is dataset-dependent, and a tailored approach to hyperparameter tuning is essential for achieving peak performance.

Sensitivity to popularity thresholds. To further validate the effectiveness of TCA4Rec in mitigating popularity bias, we conduct experiments under varying definitions of long-tail items. Specifically, we sort all items by popularity in ascending order and consider the last 50%, 60%, 70%, 80%, 90% as the long-tail subset (i.e., $r_{\text{tails}} = \{50\%, 60\%, 70\%, 80\%, 90\%\}$). Figure 7 depicts the model performance in terms of HR@5 and NDCG@5 across these thresholds. We find that the largest performance gains occur when $r_{\text{tails}} = 80\%$, indicating that our asymmetric multi-instance augmentation strategy most effectively addresses bias when focusing on the less popular 80% of items. This result underscores the robustness of TCA4Rec in handling different degrees of item popularity skew.

Overall, our experimental outcomes highlight that a careful calibration of hyperparameters-particularly the number of augmented views n and the chosen popularity threshold r_{tails}-is critical not only for improving model performance but also for enhancing convergence and stability by ensuring adequate representation of non-popular items throughout the training process.

Limitations

While TCA4Rec achieves notable performance in alleviating popularity bias and improving sequential recommendation accuracy, we acknowledge several limitations that warrant further investigation:

- **Static augmentation strategy**: The asymmetric augmentation ratios in AMINCE loss are statically defined based on item popularity distributions. This fixed scheme may not generalize well across dynamic or domain-specific user behavior patterns. Adaptive augmentation tuning based on real-time interaction feedback could offer better generalization.
- **Interpretability constraints**: Although TCA4Rec demonstrates strong empirical performance, its internal decision-making remains a black box to end users. Existing interpretability techniques such as SHAP and LIME are difficult to directly apply in sequential settings, and our attempts using t-SNE yielded limited insight. Enhanced interpretability remains an open challenge.
- **Computational overhead and deployment cost**: Despite efforts to ensure modularity and GPU compatibility, the dual-stage training and multi-instance contrastive sampling introduce non-trivial computational overhead. This may limit real-time adaptability in latency-sensitive systems. More efficient training strategies and compression techniques (e.g., knowledge distillation or pruning) should be explored in future work.

Conclusion and future work

This paper introduced TCA4Rec, a novel framework for sequential recommendation that effectively addresses the challenges of sparse supervision signals and popularity bias. By integrating an additional memory module and an Asymmetric Multi-instance Noise Contrastive Estimation (AMINCE) loss function, TCA4Rec provides robust representations of user preferences and delivers rich, bias-aware training signals. Furthermore, our two-stage training strategy ensures a balanced learning process by preventing popular items from dominating the model. Comprehensive experiments on real-world datasets demonstrate that TCA4Rec outperforms existing methods, achieving significant improvements in recommendation accuracy and robustness.

To bridge the gap toward industrial adoption, future research will: (1) Develop adaptive augmentation mechanisms (e.g., reinforcement learning, Bayesian optimization) for dynamic hyperparameter tuning; (2) Investigate model compression and distributed training to reduce inference latency and resource consumption;(3) Design dedicated interpretability frameworks tailored to sequential recommendation models.

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Author contributions
Yanan Bai designed the research idea and experiments, analyzed the experiment results, and was a major contributor in writing the manuscript. Liji Xiao and Chongjun Xia helped with the data collection, draw the experiment figures, executed the experiments. Keqiang Ze implemented the algorithm, designed and executed the experiments. Xiaoyu Shi provided general research guidance, managed the research workflow, and revised the initial versions of the manuscript. All authors read and approved the final manuscript.

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Data availibility
The datasets generated and/or analysed during the current study are available in the [Amazon product data] repository, [http://jmcauley.ucsd.edu/data/amazon/].

Declarations

Ethics approval and consent to participate
Not applicable.

Consent for publication
Not applicable.

Competing interests
The authors declare that they have no Conflict of interest.

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